

Deep Sharma

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EDUCATION

Master of Science in Computer Science - Arizona State University, Tempe, Arizona (GPA: 4.0) May 2026

- *Relevant coursework:* Cloud Computing, Data Processing at Scale, Semantic Web Mining, Advanced Algorithms

Bachelor of Technology in Computer Science Engineering - SRM Institute of Science and Technology, India (GPA: 3.5) May 2021

TECHNICAL SKILLS

Programming Languages: Java, Python, Go, C/C++, TypeScript

Cloud & DevOps: AWS, Azure, GitHub Actions, Terraform, Kubernetes, Jenkins, Ansible, Docker, Git

Web & Databases: React.js, Redux, Node.js, FastAPI, HTML/CSS, Tailwind CSS, MySQL, PostgreSQL, MongoDB

AI Tools and Frameworks: PyTorch, Hugging Face, LangChain, LlamaIndex, LangGraph, Azure OpenAI, Pinecone, MLflow, Cursor

Certifications: HashiCorp Certified: Terraform Associate (002), Azure Fundamentals (AZ-900), Azure Administrator Associate (AZ-104), Azure Developer Associate (AZ-204), DevOps Engineer Expert (AZ-400)

EXPERIENCE

Founding Software Engineer, Powerhouse9ja LLC, Phoenix, AZ, USA **Jan 2026 - Present**

- Architected and deployed Azure cloud infrastructure via Terraform, orchestrating Application Gateway, App Services, Container Services, and Functions to support rapid 0-1 scaling for a high-scale lottery platform while maintaining a 99% staging uptime.
- Designed a comprehensive data architecture layer by establishing Azure Data Lake Storage (ADLS) to capture and organize immutable draw logs, tickets, user activity, and transaction exports, alongside a base Snowflake schema synchronization pipeline.
- Optimized backend infrastructure and security design to guarantee sub 400ms ticket purchase API latency and under 2 second algorithmic winner selection, enforcing PCI DSS aligned Paystack/Flutterwave payment workflows, secure JWT token sessions, and ACID compliant wallet updates with automated webhook retry queues.

Software Engineer 2, Capgemini, Pune, India **July 2022 - July 2024**

- Developed a one-click solution using ServiceNow, Python, Terraform, and Azure DevOps to automate cloud resource provisioning, decreasing deployment time by 50%.
- Deployed Bitbucket Datacenter on AWS leveraging a Blue-Green approach, ensuring 99.99% availability, with AWS CloudFormation, Docker, RDS, and EKS (Kubernetes).
- Automated 100 EC2 spin-ups/turn-offs utilizing AWS Lambda, slashing monthly costs by approximately 40%.
- Spearheaded a 6-month Terraform remediation project, directing two developers from solution design through sprint planning to successfully clear the backlog and meet strict client timelines.
- Recognized with Capgemini Propeller Award twice (2021, 2022) for outstanding efforts in upskilling multiple teams in DevOps practices and tools.

Software Engineer, Capgemini, Pune, India **June 2021 - July 2022**

- Engineered reusable Azure and AWS Terraform modules to standardize infrastructure blueprints, cutting fresh provisioning cycles by 30% while establishing organizational best practices across cross-functional teams.
- Implemented AWS Lambda to create a ServiceNow incident whenever websites went down, alerting the Ops team to take quick action and reducing website downtime by 25%.
- Established approval gates and policies in 20+ Azure DevOps repositories, enhancing code quality and diminishing errors.
- Documented code design changes, drafted 10+ SOPs, and provided 15+ knowledge transfers (KTs) to operations teams, improving adoption of DevOps practices and infrastructure as code across organization.

Software Engineer Intern, Capgemini, Remote **Jan 2021 - June 2021**

- Engineered ETL pipelines and interactive dashboards using Azure Data Factory, Databricks, and Python (Pandas/NumPy) to efficiently ingest, process, and analyze a countrywide 49 state traffic dataset for safety trends.
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PROJECTS

Tutoring Assistant [<https://github.com/hadessharma/Tutoring-Assistant>] **May 2025 - May 2026**

- Architected an open-source, dual microservice backend using Python, FastAPI, and Gemini 1.5 Flash to decouple low-latency student workflows from data administration while enforcing strict Socratic guardrails.
 - Engineered a course-isolated RAG pipeline leveraging LangChain and Supabase (pgvector) to ingest and retrieve materials, partitioning data by course code to guarantee exact contextual accuracy.
 - Built the frontend using Next.js, React, and Tailwind CSS with an automated engine that compiles unstructured student dialogue logs and learning gaps into structured topic by topic study plans.
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LEADERSHIP EXPERIENCE

Senior Tutoring Shift Lead, ASU Engineering, Tempe, Arizona, USA **Feb 2025 - May 2026**

- Awarded the **2026 Best Student Leader Award** and nominated for national recognition (NSEA) for scaling tutoring operations, designing comprehensive training modules, and managing scheduling/payroll for a staff of 100 employees.
- Led the incorporation of AI for tutoring and utilized agentic coding to architect and deploy engaging, collaborative gaming and educational websites, enhancing digital interaction for the engineering student body.